Burak Satar

Research Scientist, Singapore Management University

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I build benchmarks and methods that test whether vision-language models understand a cultural scene or only sound confident about it, starting with Southeast Asia. Lead author of Cultural Moment (EMNLP 2026, oral) and co-first author of Seeing Culture (EMNLP 2025).

EXPERIENCE

Research Scientist, Singapore Management University (SMU), Singapore — Mar 2025 – Present
Making vision-language models culturally aware, with reasoning across image, video, audio and text, starting with Southeast Asia. Working with Prof Chong-Wah Ngo.

Doctoral Researcher (SINGA Scholar), Institute for Infocomm Research (I²R), A*STAR, Singapore — Jan 2020 – Dec 2024
Semantic, debiased and moment-level text-to-video retrieval with multi-modal features.

Visiting Researcher, MaVi Research Group, University of Bristol, UK — Oct 2023 – Dec 2023
Video corpus moment retrieval, with Dr Michael Wray in Dima Damen's group.

Machine Learning Engineer, Turkish Technic (Turkish Airlines), İstanbul, Türkiye — Feb 2019 – Sep 2019
Drone-based aircraft inspection with computer vision for automated defect recognition; cut total maintenance time by over 30%.

Software Engineer (Erasmus+ traineeship), University of Valencia, Spain — Sep 2016 – Mar 2017
IoT sensor-data dashboard (MQTT, Node.js) in the Digital Signal Processing Lab.

Electronics Engineer, TSV Energy (start-up), Türkiye — Dec 2014 – Sep 2015
Wireless sensor network and Linux-based hub for a smart-home product.

Machine Learning Advisor, two award-winning start-ups in London and İstanbul

EDUCATION

Ph.D. in Computer Science, Nanyang Technological University (NTU), Singapore — Jan 2020 – Dec 2024
Thesis: *Towards Semantic, Debiased and Moment Video Retrieval with Multi-modal Features*. Supervised by Prof Joo-Hwee Lim, Dr Hongyuan Zhu and Prof Hanwang Zhang. A*STAR SINGA scholarship. CGPA 4.25/5.00.

M.Sc. in Electronics Engineering, Bursa Uludağ University, Türkiye — Sep 2015 – Aug 2018
Thesis on vehicle detection and make-model classification with deep learning, supervised by Prof Ahmet Emir Dirik.

B.Sc. in Electronics Engineering, Bursa Uludağ University, Türkiye — Sep 2009 – Nov 2014
Exchange semesters in Siena and Naples, Italy.

PUBLICATIONS

Conference papers

1. Cultural Moment Benchmark: Evaluating Video Cultural Reasoning and Grounding in Southeast Asia
   Burak Satar, Zhixin Ma, Cheng Yu-Tong, Huy Hoang Tran, Phuong Anh Nguyen, Chong-Wah Ngo. *EMNLP 2026 (Main Conference, oral)*.
2. Seeing Culture: A Benchmark for Visual Reasoning and Grounding
   Burak Satar, Zhixin Ma, Patrick Amadeus Irawan, Wilfried Ariel Mulyawan, Jing Jiang, Ee-Peng Lim, Chong-Wah Ngo. *EMNLP 2025 (Main Conference)*.
3. VG-TVP: Multimodal Procedural Planning via Visually Grounded Text-Video Prompting
   Muhammet Furkan Ilaslan, Ali Köksal, Kevin Qinghong Lin, **Burak Satar**, Mike Zheng Shou, Qianli Xu. *AAAI 2025*.
4. Towards Debiasing Frame Length Bias in Text-Video Retrieval via Causal Intervention
   Burak Satar, Hongyuan Zhu, Hanwang Zhang, Joo-Hwee Lim. *BMVC 2023*.
5. Semantic Role Aware Correlation Transformer for Text to Video Retrieval
   Burak Satar, Hongyuan Zhu, Xavier Bresson, Joo-Hwee Lim. *IEEE ICIP 2021 (oral); also presented at the ICCV 2021 SRVU Workshop (oral)*.
6. Deep Learning Based Vehicle Make-Model Classification
   Burak Satar, Ahmet Emir Dirik. *ICANN 2018 (oral)*.

Workshop papers and preprints

7. An Overview of Challenges in Egocentric Text-Video Retrieval
   Burak Satar, Hongyuan Zhu, Hanwang Zhang, Joo-Hwee Lim. *CVPR 2023, Joint Ego4D/EPIC Workshop (extended abstract, oral)*.

8. RoME: Role-aware Mixture-of-Expert Transformer for Text-to-Video Retrieval
   Burak Satar, Hongyuan Zhu, Hanwang Zhang, Joo-Hwee Lim. *arXiv preprint, 2022*.
9. Exploiting Semantic Role Contextualized Video Features for Multi-Instance Text-Video Retrieval (EPIC-KITCHENS-100 Challenge 2022)
   Burak Satar, Hongyuan Zhu, Hanwang Zhang, Joo-Hwee Lim. *CVPR 2022, Joint Ego4D/EPIC Workshop (challenge technical report, joint 3rd place)*.

Under review and in preparation

10. Cultural Video Corpus Moment Retrieval
    Burak Satar, Zhixin Ma, Chong-Wah Ngo. *ACL 2027 (under review)*.
11. Video Object Segmentation in Cultural Context
    Zhixin Ma, **Burak Satar**, Chong-Wah Ngo. *CVPR 2027 (under review)*.
12. Visually Grounded Augmentation for Reasoning Segmentation of Cultural Artifacts
    Zhixin Ma*, **Burak Satar***, Phuong Anh Nguyen, Chong-Wah Ngo. *ACM TOMM (under review)*. * equal contribution
13. Enhancing Video Corpus Moment Retrieval in Long Ego-centric Videos with LLM and Audio Fusion
    Burak Satar, Joo-Hwee Lim, Hanwang Zhang, Muhammet Furkan Ilaslan, Hongyuan Zhu, Michael Wray. *In preparation*.

AWARDS AND HONOURS

- **Oral presentation**, EMNLP 2026 Main Conference (2.7% acceptance rate)
- **Joint 3rd place**, EPIC-Kitchens-100 Multi-Instance Retrieval Challenge, CVPR 2022
- **Finalist**, Three Minute Thesis (3MT), NTU, representing CCDS, 2022
- **SINGA Ph.D. Scholarship**, A*STAR, 2020–2024
- **Student Travel Award**, European Neural Network Society (ICANN), 2018

TALKS

- **Developing Research Skills: From Process to Paper**, CCDS Graduate Students' Club, NTU, Mar 2026 (invited)
- **Seeing Culture**, ResearchTrend.AI, Dec 2025 (invited)
- **Orals**: EMNLP 2026, CVPR 2023 Joint Ego4D/EPIC Workshop, ICCV 2021 SRVU Workshop, IEEE ICIP 2021, ICANN 2018
- **Posters**: BMVC 2023, Singapore Vision Day 2023

ACADEMIC SERVICE

- **Workshop bid**: a workshop on cultural understanding at CVPR 2027 and ICMR 2027, with AI Singapore, MBZUAI and Oracle
- **Registration Chair**, SSNLP 2026
- **Programme Committee and reviewer**: AAAI 2026 and 2027, ACM MM 2026, ICANN 2019, IEEE Transactions on Artificial Intelligence
- **Judge**, Singapore Science and Engineering Fair (SSEF) 2025
- **Volunteer**: ACM Web Conference 2024, NeurIPS 2021
- **Summer schools**: CIFAR DLRL 2022; PAISS 2021 (poster)

LEADERSHIP AND COMMUNITY

- **Co-founder**, Turquoise Dot, a community for Turkish and Turkic researchers and tech professionals in Singapore, 2025–Present
- **Co-founder and President**, NTU Turkish Society, 2021–2024
- **Member**, NTU Graduate Students' Association Welfare Committee, 2021–2024

SKILLS AND REFERENCES

Research	Vision-language and multimodal LLMs; cultural reasoning and grounding; text-video and moment retrieval; benchmark and dataset construction; causal inference
Engineering	Python, PyTorch, scikit-learn, pandas; training and evaluating multimodal models at scale; Linux
Languages	Turkish (native), English (professional working proficiency), Italian, Spanish and Bahasa Indonesia (elementary)
Memberships	IEEE, CVF, ACM, ACL, PREMIA
References	Prof Chong-Wah Ngo (SMU) · Prof Joo-Hwee Lim (A*STAR, NTU) · Dr Hongyuan Zhu (A*STAR) · Prof Hanwang Zhang (NTU)